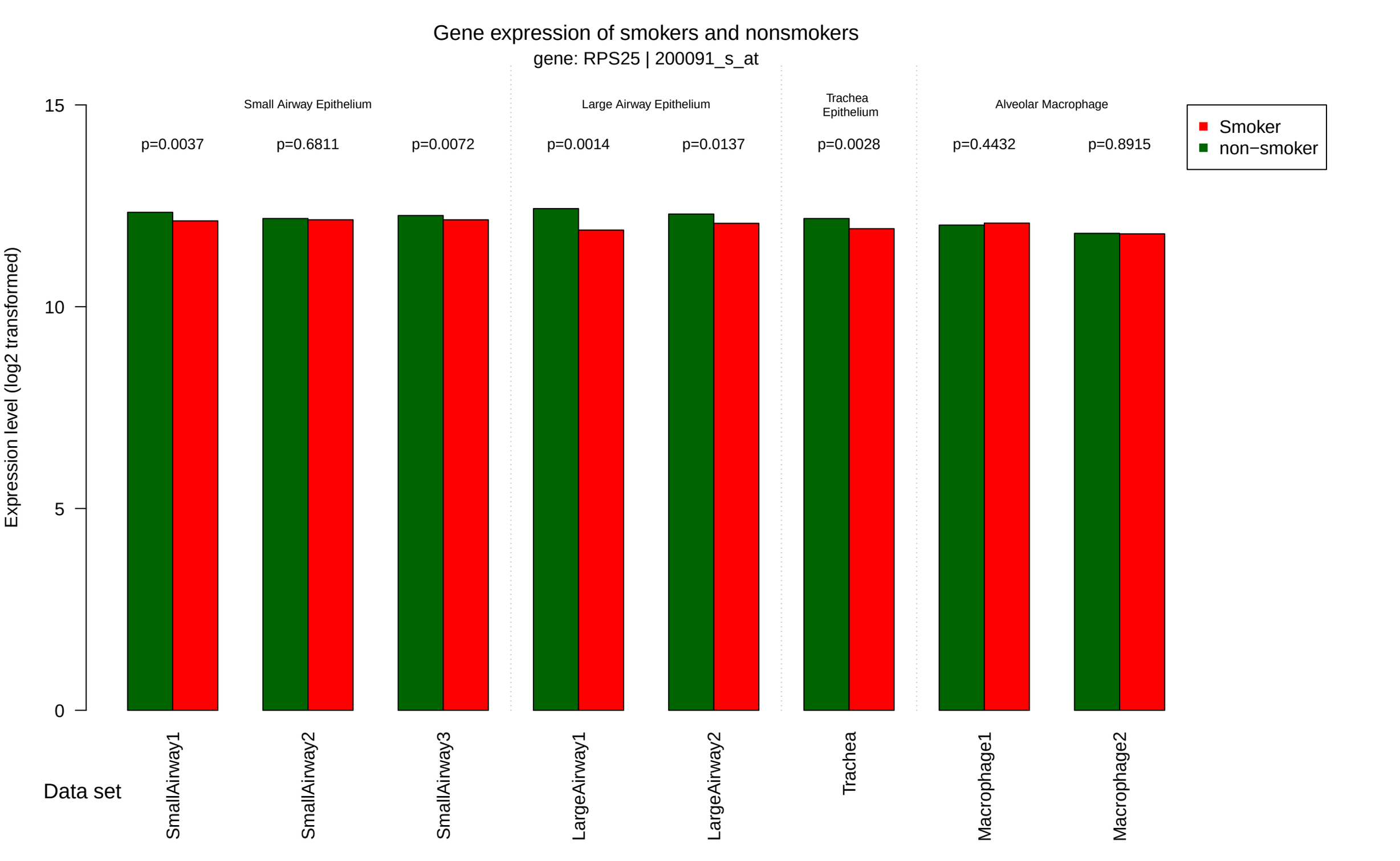
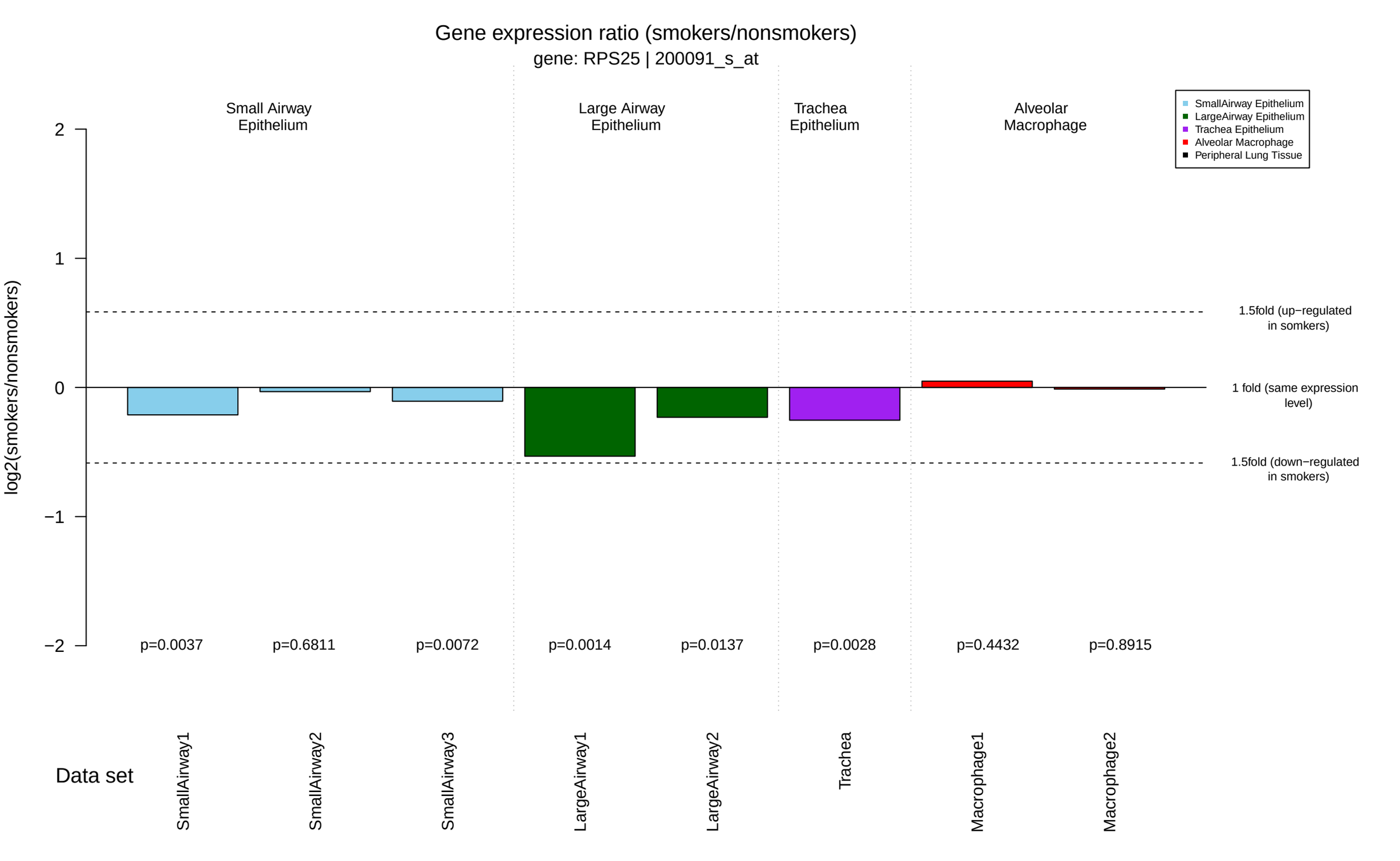
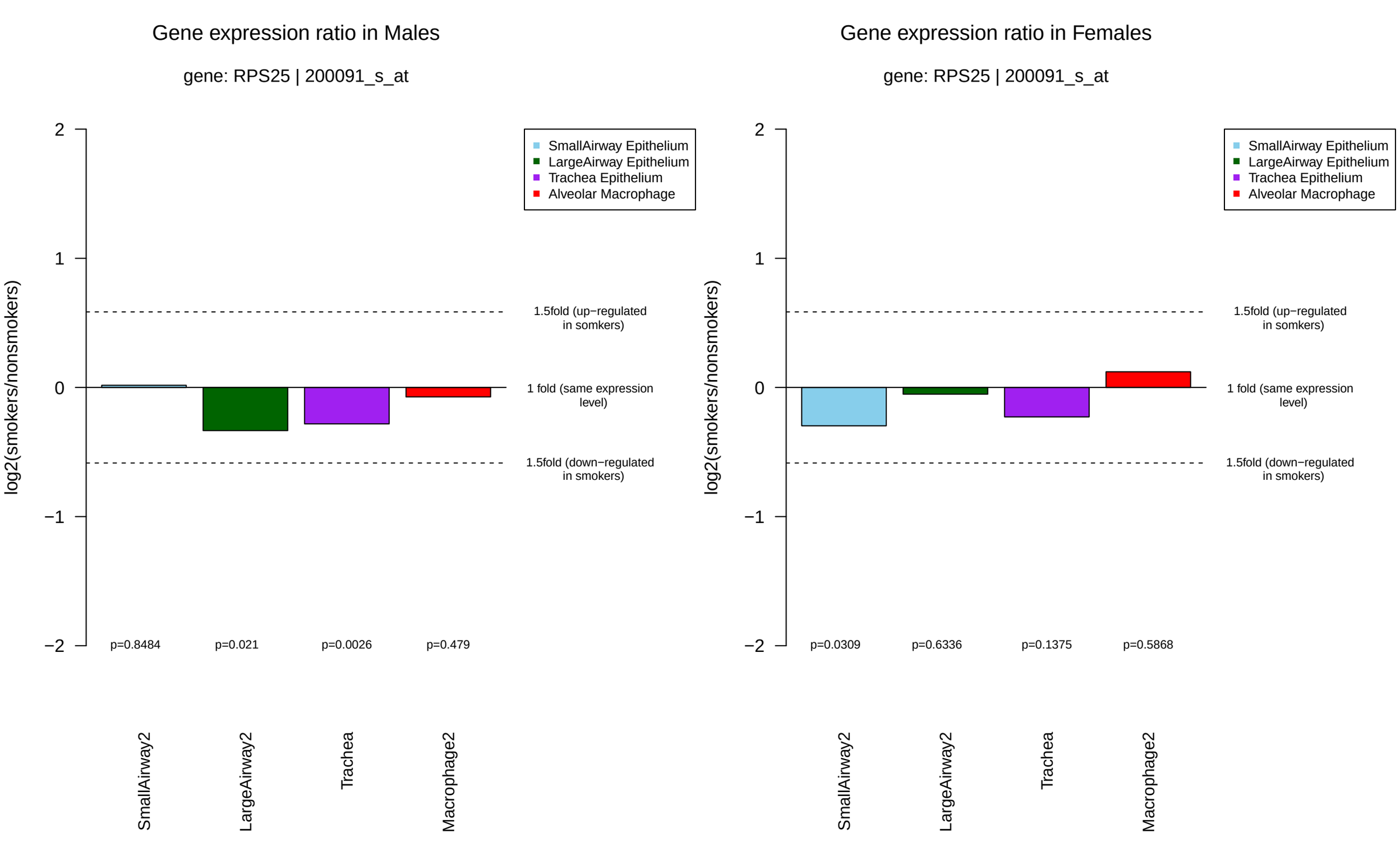
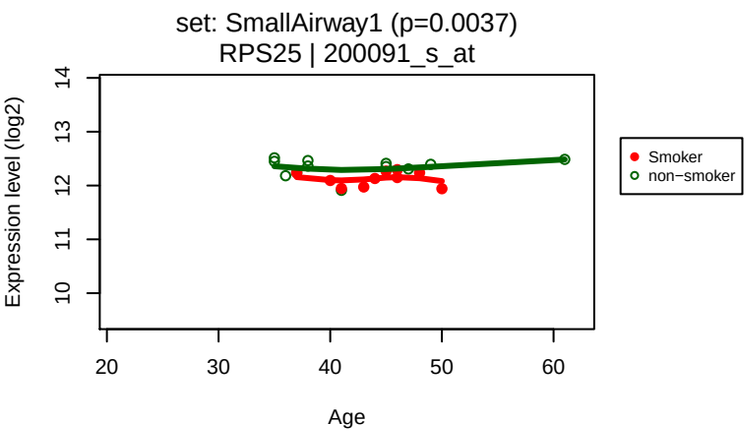
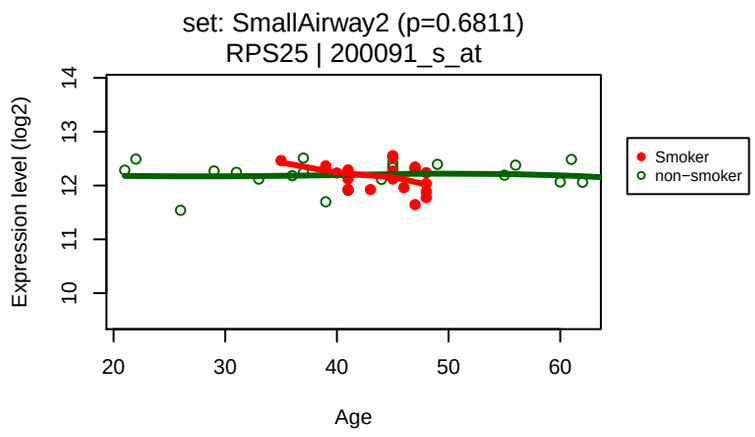
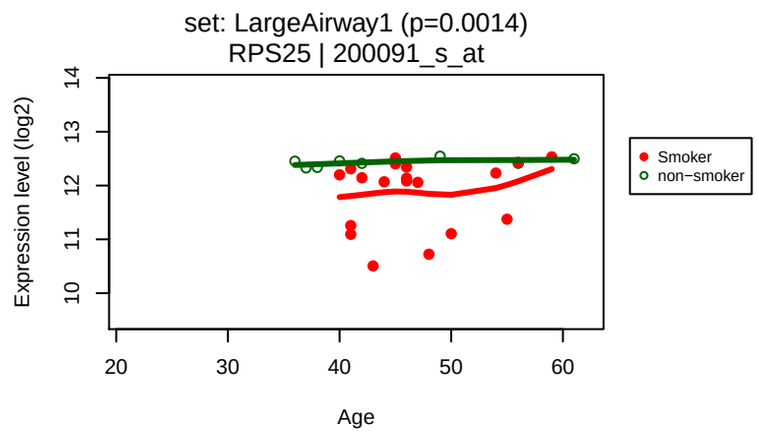

Gene expression of smokers and nonsmokers
gene: RPS25 | 200091_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0037
p=0.6811
p=0.0072
p=0.0014
p=0.0137
p=0.0028
p=0.4432
p=0.8915
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

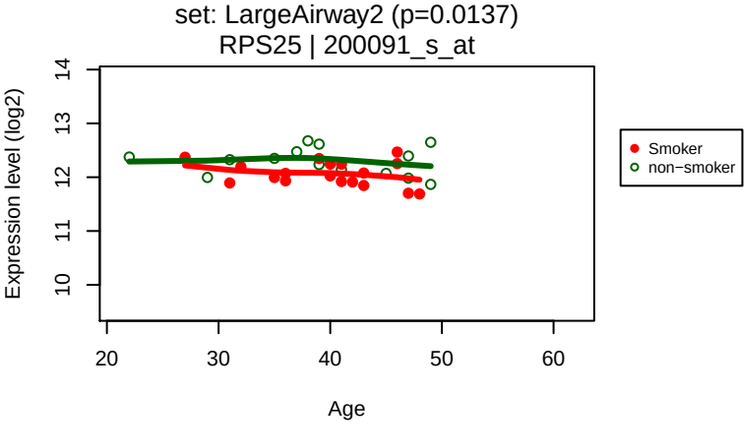

Gene expression ratio (smokers/nonsmokers)
gene: RPS25 | 200091_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0037
p=0.6811
p=0.0072
p=0.0014
p=0.0137
p=0.0028
p=0.4432
p=0.8915
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

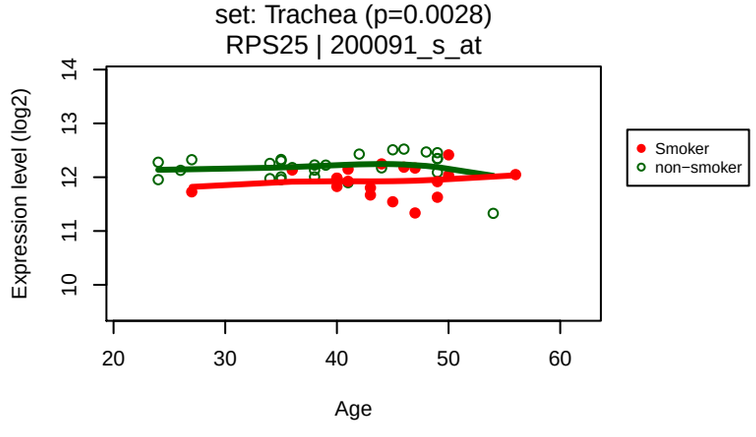

Gene expression ratio in Males
gene: RPS25 | 200091_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8484
p=0.021
p=0.0026
p=0.479
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS25 | 200091_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0309
p=0.6336
p=0.1375
p=0.5868
SmallAirway2
LargeAirway2
Trachea
Macrophage2

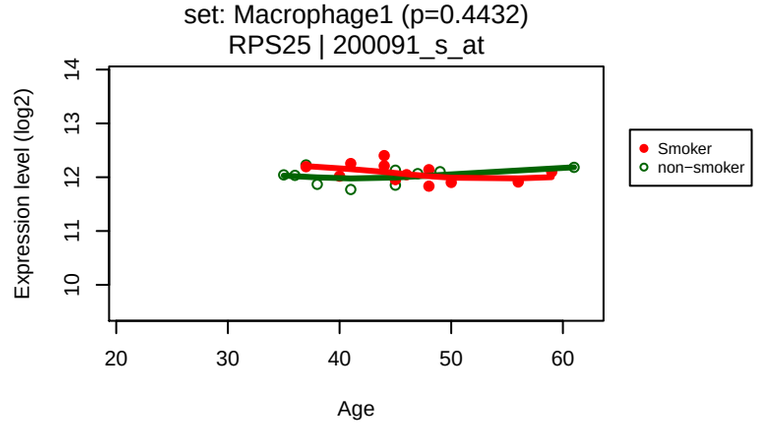

set: SmallAirway1 (p=0.0037)
RPS25 | 200091_s_at
Expression level (log2)
Age
Smoker
non−smoker

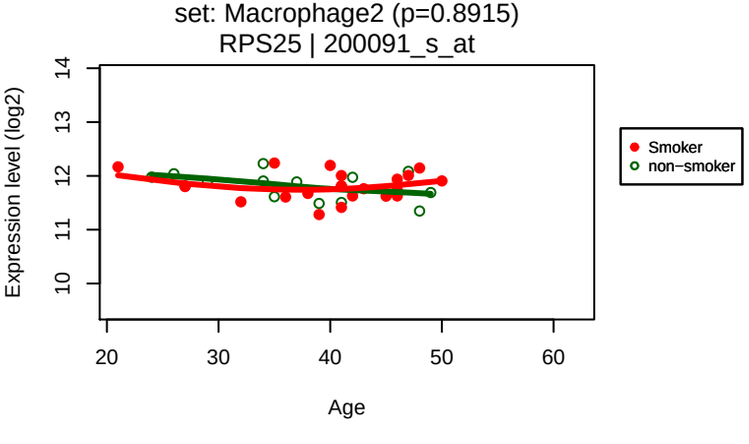

set: SmallAirway2 (p=0.6811)
RPS25 | 200091_s_at
Expression level (log2)
Age
Smoker
non−smoker

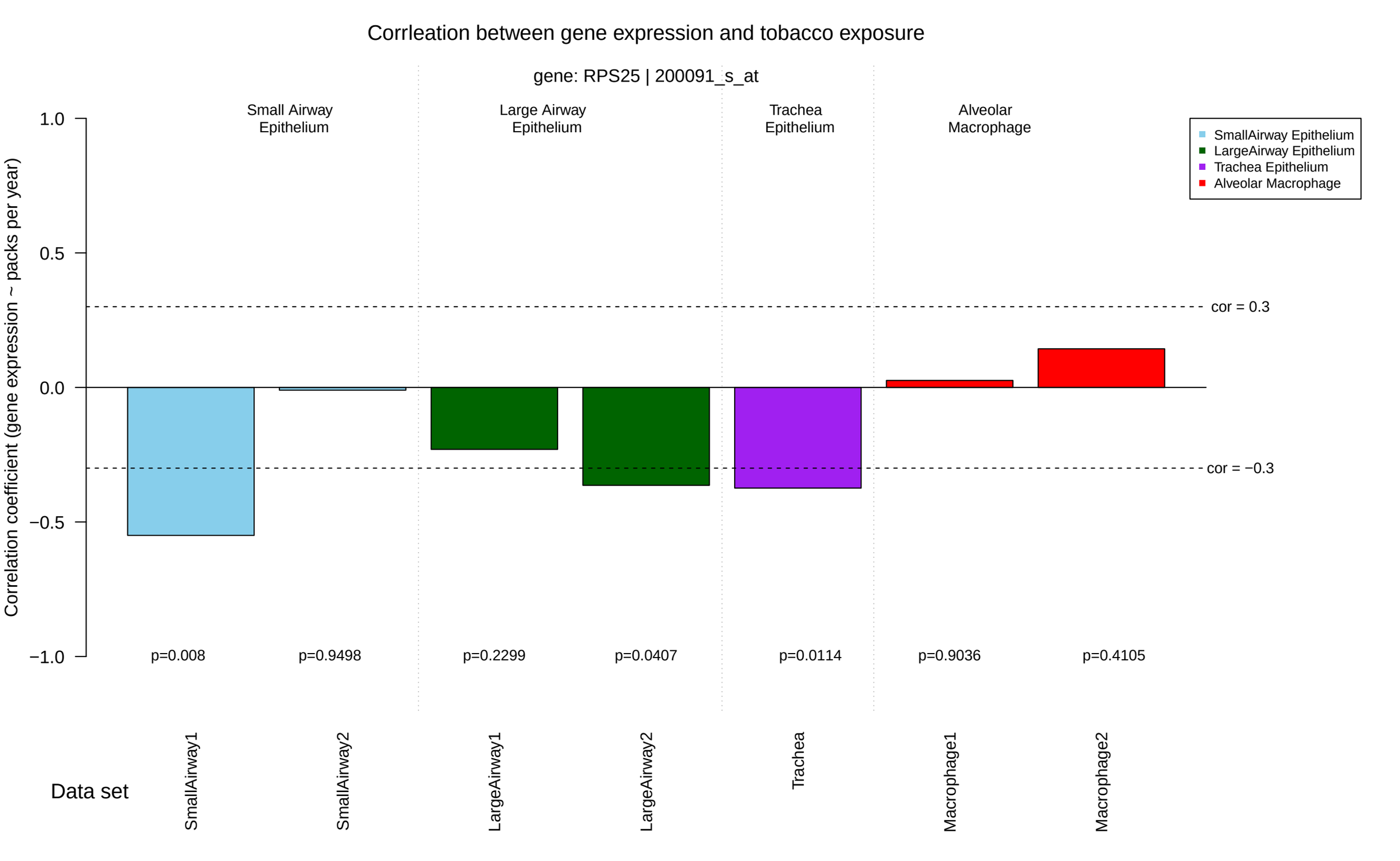

set: LargeAirway1 (p=0.0014)
RPS25 | 200091_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0137)
RPS25 | 200091_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0028)
RPS25 | 200091_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4432)
RPS25 | 200091_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8915)
RPS25 | 200091_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS25 | 200091_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.008
p=0.9498
p=0.2299
p=0.0407
p=0.0114
p=0.9036
p=0.4105
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2